HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python**-based data pipelines on **AWS Lambda** to process flight offer data, handling approximately **3,500 events per second** during peak load.
- Developed **Java Spring Boot** microservices for booking servicing, managing changes, cancellations, and refunds across airline and supplier systems, ensuring consistent workflow execution.
- Engineered **REST APIs** using **Java Spring Boot** to abstract complex flight search and booking logic, simplifying partner integrations and reducing the need for deep airline domain knowledge.
- Contributed to the design of **API architecture** and **data models** for flight search and booking services, focusing on scalability and maintainability within a small, hands-on engineering team.
- Implemented **caching strategies** for flight offers using **Redis**, balancing retrieval speed, cost, price accuracy, and data freshness to optimize performance for high-traffic systems.
- Supported **modern airline retailing standards**, integrating **New Distribution Capability (NDC)** workflows with existing Global Distribution System (GDS) interactions to enhance flight offer availability.
- Containerized backend services with **Docker** and deployed to an **AWS EKS** cluster, configuring basic resource limits and monitoring within a CI/CD pipeline managed by **Jenkins**.

SKILLS

Languages – Python, Java, SQL, R, Bash

Frameworks/Libraries – Spring Boot, Hibernate, JPA, pandas, NumPy, scikit-learn, FastAPI, Flask

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch, EKS)

Databases – PostgreSQL, MySQL, Oracle, Redis

Tools & Technologies – Docker, Kubernetes, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace, Power BI, Tableau, GDS, NDC

Methodologies – Agile, Scrum, CI/CD

PROJECTS

Scalable Flight Offer API (HCL Technologies)

- Developed a **Python**-based microservice using **FastAPI** to serve flight offer data, integrating with a PostgreSQL database for real-time availability checks.
- Designed and implemented caching layers with **Redis** to improve read performance by **40%** for frequently requested flight routes, reducing database load.
- Wrote comprehensive unit and integration tests using **pytest**, achieving over **90% code coverage** for core API endpoints.

Flight Booking Servicing Module (HCL Technologies)

- Built a **Java Spring Boot** application module to handle post-booking operations including modifications and cancellations, interacting with simulated airline backend APIs.
- Integrated with a mock **GDS** interface to simulate transaction flows for booking changes, ensuring data consistency across systems.
- Implemented **JPA** for data persistence, managing entity relationships for bookings, customer details, and transaction logs.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)